



Steps:

1. Create an origin web server on a platform of your choosing. This could be in AWS, Google Cloud, DigitalOcean, your Raspberry Pi, etc.

This web server must run an endpoint that returns all HTTP request headers in the body of the HTTP response.

The web server can be something that you have written yourself (e.g. in JavaScript, Python, etc) or by using a 3rd party application.

2. Proxy traffic to this server through Cloudflare. Add necessary configurations on Cloudflare.
3. Secure the communication between Cloudflare and your Origin Server with a non-Cloudflare provisioned TLS certificate using at least the Full-Strict mode on Cloudflare.
4. Install and configure Cloudflare Tunnel on your origin server using a subdomain called "tunnel", e.g. tunnel.yourwebsite.com. Make connections proxied to your server protected using this tunnel.
5. Configure an SSO IdP (Identity Provider) of your choosing within **Cloudflare Zero Trust**.
6. Lock down access for a particular path for your Cloudflare Tunnel subdomain (e.g. tunnel.yourwebsite.com/secure) and only allow access for yourself (with the previously configured IdP of your choosing) and users with an @cloudflare.com email address .
 - a. Ensure nobody can bypass Cloudflare and access your server's IP directly.

Please let us know if you have any questions with any of the above!

Reference links:

<https://developers.cloudflare.com/fundamentals/setup/account-setup/add-site/>

<https://developers.cloudflare.com/ssl/get-started/>

<https://developers.cloudflare.com/learning-paths/get-started/>

<https://developers.cloudflare.com/cloudflare-one/connections/connect-networks/>

<https://developers.cloudflare.com/cloudflare-one/identity/idp-integration/>